Activity 1

PROJECT PROPOSAL

Jaffna EcoGrow – Food-Waste-Based Organic Fertilizer Business

Activity 1: Small Business Proposal Based on a Local Natural Resource

1. Introduction

Jaffna EcoGrow is a proposed small-scale business that converts locally available biodegradable food waste into organic fertilizer. The business will collect suitable organic waste from selected markets, restaurants, hotels, fruit and vegetable shops, and other food-related businesses in the Jaffna area. The waste will be sorted and processed using controlled composting methods to produce a useful organic fertilizer/soil amendment for agricultural and gardening purposes.

2. Background and Problem Statement

Food and vegetable waste is produced every day by households, markets, restaurants, and food businesses. When organic waste is poorly managed, it can create unpleasant odours, attract pests, and increase the amount of waste requiring disposal. At the same time, farmers and gardeners need affordable inputs that can improve soil quality. Jaffna EcoGrow proposes to address both issues by transforming biodegradable food waste into a useful agricultural product.

3. Natural Resource Available in Jaffna

The main resource for the proposed business is locally available biodegradable food and agricultural waste. Examples include fruit and vegetable residues, fruit peels, rejected produce, and other suitable organic materials. These materials are treated as a resource rather than simply as waste because their organic matter can be converted into compost and soil-improving products.

4. Proposed Business Idea

The business will establish a local collection and processing system. Organic waste will be collected from selected suppliers, inspected and separated to remove plastics, glass, metals, and other unsuitable materials. The suitable organic material will then undergo controlled composting and maturation. After quality checks, the finished product will be screened, packaged, labelled, and sold under the Jaffna EcoGrow brand.

5. Objectives

• Reduce the amount of biodegradable food waste requiring disposal.
• Convert low-value organic waste into a marketable agricultural product.
• Supply locally produced organic fertilizer to farmers and gardeners.
• Create employment and income opportunities in the Jaffna area.

• Promote environmentally responsible waste management and a circular-economy approach.

6. Main Product

The initial product will be Jaffna EcoGrow Organic Fertilizer, produced from properly processed biodegradable food waste. It can be offered in different package sizes for home gardeners, nurseries, and farmers. In later stages, the business could develop crop-specific or soil-specific formulations, subject to appropriate testing and regulatory requirements.

7. Target Market

The initial target customers will include small and medium-scale farmers, home gardeners, plant nurseries, landscaping businesses, and other agricultural users in Jaffna and nearby areas. Direct sales to farms and partnerships with nurseries and agricultural retailers can be used to reach customers efficiently.

8. Production Process

The proposed production process is:

Food-waste suppliers → Collection → Sorting → Shredding/mixing → Controlled composting → Maturation → Screening → Quality checking → Packaging → Distribution

Good waste segregation at the source will be important because contamination with plastic or other non-biodegradable materials can reduce product quality and increase processing costs.

9. Monopoly Market Structure

For the economics activity, the business will be considered within a narrowly defined local market, rather than claiming a monopoly over all fertilizer sales in Jaffna. The proposed market is specialized organic fertilizer produced from locally collected food waste within the selected service area. The business could seek exclusive long-term agreements with selected food markets, restaurants, hotels, and food-processing businesses for collection of their organic waste. It could also develop specialized
processing know-how and a dedicated local distribution network. These arrangements can create barriers to entry for potential competitors in the defined market.

10. Barriers to Entry

Potential barriers to entry include exclusive waste-collection agreements, the cost of establishing a suitable composting and processing facility, the need for reliable waste-sorting and quality-control systems, established relationships with local customers, and the development of specialized product formulations and distribution channels. These barriers do not prevent all

competition in the wider fertilizer market, but they can make entry into the specific proposed market more difficult.

11. Pricing and Revenue

The business will generate revenue mainly through the sale of packaged organic fertilizer. Pricing will consider production and distribution costs, customer demand, the prices of substitute fertilizers, product quality, and the required profit margin. Because customers may have alternative fertilizer products, the business cannot increase its price without considering the resulting effect on demand.

12. Main Costs

Major costs are expected to include waste collection and transportation, land or facility costs, composting equipment, labour, water and electricity, packaging materials, quality testing, marketing, maintenance, and distribution. A potential cost advantage is that suitable food waste may be obtained at low cost through collection agreements, although collection and transportation still create operating expenses.

13. Marketing Strategy

The business will emphasize three main benefits: locally produced organic fertilizer, conversion of food waste into useful products, and support for sustainable agriculture. Marketing can be carried out through local agricultural communities, nurseries, social media, demonstrations, sample distribution, and partnerships with retailers. Customer feedback can be used to improve packaging, product consistency, and service.

14. Environmental and Social Benefits

The proposed business can reduce the amount of biodegradable waste requiring disposal and create value from materials that would otherwise have little economic value. It can also create local employment, provide an additional market for waste collection services, and support agricultural activities by supplying an organic soil-improving product.

15. Risks and Challenges

Important challenges include inconsistent waste supply, contamination of collected waste, odour and pest management, maintaining consistent product quality, transportation costs, customer acceptance, competition from existing fertilizer products, and compliance with relevant environmental and agricultural requirements. These risks can be reduced through source segregation, controlled processing, quality monitoring, staff training, and proper facility management.

16. Conclusion

Jaffna EcoGrow proposes a practical waste-to-value business that uses locally available biodegradable food waste as its main resource. By converting organic waste into a marketable fertilizer product, the business can create economic and environmental value. For the economics activity, its monopoly characteristics are considered within a clearly defined local niche market, supported by potential barriers such as exclusive supply relationships, specialized processing capability, and an established distribution network.

Activity 2

Activity 2 – Monopoly Market Failure

Case Study: Coal India Limited

1. Introduction

A monopoly or dominant market position can reduce competition and give a firm greater control over market conditions. When a dominant firm abuses its market power, it can lead to market failure and negatively affect other businesses and consumers.

This case study examines **Coal India Limited (CIL)** and its subsidiaries in India's non-coking coal market. The Competition Commission of India (CCI) found that Coal India held a dominant position in the relevant market and that certain practices amounted to an abuse of this position.

2. Background of Coal India

Coal India Limited is a major coal-producing company in India. Through its subsidiaries, it supplies coal to different industries, particularly power producers.

Coal is an important input for electricity generation. Therefore, power producers depend on a reliable supply of coal for their operations.

According to the CCI, Coal India and its subsidiaries had a dominant position in the relevant market for the production and supply of non-coking coal in India.

3. Why Was Coal India Dominant?

Coal India had a very strong position in the production and supply of non-coking coal. Buyers had limited alternative sources compared with a competitive market.

As a result, Coal India had greater bargaining power when dealing with some customers. The CCI noted that its market position allowed it to operate with limited competitive pressure in the relevant market.

However, it is important to note that **having a dominant position itself is not illegal**. The competition-law concern arises when that position is abused

4. The Fuel Supply Agreement Problem

Coal India used **Fuel Supply Agreements (FSAs)** with its customers. These agreements contained conditions relating to:

- Quantity of coal
- Coal quality and grade
- Sampling and testing
- Transportation
- Compensation
- Delivery conditions

The CCI found that certain FSA conditions were **unfair and discriminatory**. It also found that some terms were imposed without a genuine bilateral negotiation process.

This was a problem because buyers with limited alternatives had less bargaining power to negotiate better conditions.

5. How Did This Cause Market Failure?

The market failure can be explained as follows:

Dominant position of Coal India
↓
Limited alternatives for buyers
↓
Weak bargaining power of buyers
↓
Unfair or discriminatory conditions
↓
Additional costs and uncertainty
↓
Reduced competitive pressure
↓
Less efficient market outcome

In a competitive market, buyers can change suppliers if they are dissatisfied with prices, quality, or contract conditions. This creates pressure on suppliers to provide reasonable terms.

In this case, the limited alternatives available to some buyers reduced this competitive pressure. Therefore, the abuse of market power contributed to an inefficient and unfair market situation.

6. Impact on the Market

Impact on Buyers

Buyers had lower bargaining power and could face uncertainty regarding coal quality, testing, transportation costs, and contractual conditions.

Impact on Competition

The strong position of Coal India reduced competitive pressure in the relevant market.

Impact on Businesses

Power producers and other industries using coal could experience additional costs or uncertainty because coal is an important input for their operations.

Impact on Market Efficiency

When competition is weakened, prices, quality, and contractual conditions may not be determined as efficiently as they would be in a competitive market.

7. Solutions I Would Propose

If I were the decision maker, I would propose the following solutions:

1. Encourage Competition

Encourage alternative suppliers and sources of coal where economically and legally possible. This would give buyers more choices.

2. Ensure Fair Contracts

Fuel Supply Agreements should contain clear, transparent, and balanced conditions, with meaningful negotiation between both parties.

3. Independent Quality Testing

Independent or standardized coal sampling and testing should be introduced to reduce disputes about coal quality.

4. Stronger Regulation

Competition authorities should regularly monitor dominant companies and investigate possible abuse of market power.

5. Buyer Complaint Mechanism

Buyers should have an effective system for reporting unfair contractual conditions and resolving disputes.

8. Action Taken by the CCI

The Competition Commission of India took action against Coal India and its subsidiaries. CCI directed them to **cease the abusive conduct and modify the relevant FSA clauses**. The modifications included areas such as sampling and testing, transportation-related expenses, and compensation provisions.

The relevant proceedings also included a financial penalty imposed on Coal India

9. Conclusion

The Coal India case demonstrates how a dominant market position can contribute to market failure when it is abused. Coal India's large market position gave it significant bargaining power, while some buyers had limited alternatives. The CCI found certain FSA conditions to be unfair and discriminatory.

The case shows the importance of **competition, fair contractual agreements, transparent quality testing, and effective regulation** in maintaining an efficient market.

References

1. Competition Commission of India – *Antitrust: Abuse of Dominant Position*.
2. Competition Commission of India – *Annual Report 2016–17*.
3. Competition Commission of India – *Fair Play Newsletter*.